higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

SYSTEM ANALYSIS AND DESIGN
NQF LEVEL 3

4 MARCH 2019

This marking guideline consists of 9 pages.

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SECTION A

QUESTION 1

1.1	1.1.1	D		
	1.1.2	D		
	1.1.3	B		
	1.1.4	A		
	1.1.5	D		
	1.1.6	B		
	1.1.7	C		
	1.1.8	D		
	1.1.9	D		
	1.1.10	A		
			(10 × 1)	(10)
1.2	1.2.1	False		
	1.2.2	False		
	1.2.3	True		
	1.2.4	True		
	1.2.5	True		
	1.2.6	True		
	1.2.7	False		
	1.2.8	True		
	1.2.9	True		
	1.2.10	True		
			(10 × 1)	(10)
1.3	1.3.1	Request for quotation		
	1.3.2	Business process automation		
	1.3.3	Office automation system		
	1.3.4	Database management system		
	1.3.5	Business process improvement		
	1.3.6	Business process re-engineering		
	1.3.7	Return on investment		
	1.3.8	Strengths, weaknesses, opportunities, threats		
	1.3.9	Executive information systems		
	1.3.10	Wide area network		
			(10 × 1)	(10)
1.4	1.4.1	accounting system		
	1.4.2	Rental		
	1.4.3	chief information officer		
	1.4.4	Information technology policies		
	1.4.5	Benchmarking		
	1.4.6	Organisational planning		
	1.4.7	Ergonomics		
	1.4.8	chief technology officer		

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1.4.9 Decision support systems
1.4.10 system

(10 × 1) (10)
[40]

TOTAL SECTION A: 40

SECTION B

QUESTION 2

2.1
- Structured analysis
- Object-oriented analysis
- Information engineering
- Rapid application development
- Joint application development
- Agile development (6)

2.2
- Preliminary investigation or feasibility analysis
- Analysis and requirements determination
- Design and requirements specification
- Coding and development
- Testing and installation
- Maintenance and support
- System review (7)

2.3
- To solve a business problem
- To be able to capitalise on an opportunity
- To comply with a directive or law (Any 2 × 1) (2)

[15]

QUESTION 3

3.1 Systematic approach to transforming raw data into information using a predefined sequence of operations (2)

3.2
- Information systems controls
- Procedural controls
- Information systems facility controls
- Information systems audits (4)

3.3
- A transaction processing system is an information system that captures and processes data of day-to-day business activities.
- A decision support system is an interactive software-based system intended to help decision makers compile useful information from raw data, documents, personal knowledge and/or business models to identify and solve problems and make decisions.
- A knowledge management system is a distributed hypermedia system for managing, storing and manipulating knowledge in organisations, using knowledge extraction tools and technologies. (3 × 2) (6)

3.4
- An accounting information system automates the role of accounting duties and functions of accountants in an organisation.
- A total information system provides centralised, integrated software to help manage and co-ordinate the ongoing activities of the enterprise, including manufacturing and distribution, accounting, finance, sales, product planning and human resources. (2 × 2) (4)

[16]

QUESTION 4

4.1 The managing of information systems in order to contribute to a company's strategic objectives (2)

4.2 4.2.1 Responsible for internal and external networks

4.2.2 Responsible for putting all information related systems in line with organisational needs

4.2.3 Establishes a culture of knowledge and knowledge management infrastructure in the organisation

4.2.4 Manages emerging technologies in terms of research, planning and adoption and is responsible for information systems architecture

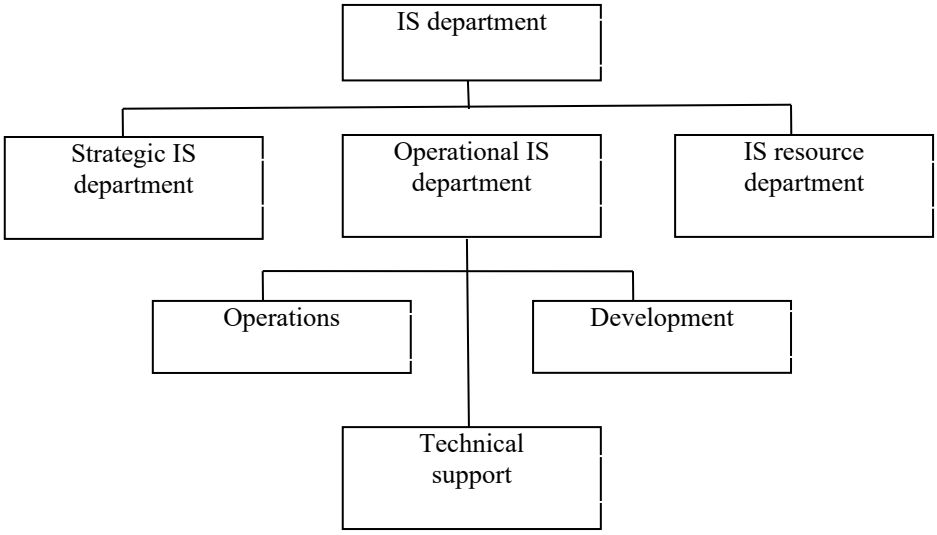
(4 × 2) (8)

4.3
- Project planning, control and management
- Systems analysis
- Systems design
- Systems coding, testing and implementation
- Quality assurance, systems reviews and acceptance testing
- End-user training and support
- Systems maintenance and enhancement (7)

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4.4
- Centralisation refers to the process whereby units or departments are centralised or put under one control.
- Decentralisation refers to the distribution of units or departments so that the components or departments can be accessed from their distributed sites. (2 × 2) (4)

4.5



(7)
[28]

QUESTION 5

5.1 Advantages
- Allows for creative and flexible solutions to business problem
- Can match business requirements exactly as stated
- Develops staff knowledge and skills for maintenance and support

Disadvantages
- Requires high level of requirement resources
- Usually has a longer development time
- High risk in terms of project success
- Can run overtime and exceed the budget (Any 3 × 2) (6)

5.2 Organisations adopt standards to guide the development of new systems or the acquisition of new computer technology due to reasons of reducing maintenance support, reducing complexity, volume discounts for purchasing and support, et cetera. (3)

5.3
- Physical space
- Ergonomics
- Security
- Power
- Network connectivity (5)

5.4
- Information systems outsourcing is an arrangement in which one company provides services or software development for another company that could also be or usually have been provided in-house.
- Information systems off-shoring means a company moves business processes or services to a country other than its home country or primary marketplace. (2 × 2) (4)

5.5
- Specify the information systems technology requirements.
- Create and issue a request for a proposal or a quotation.
- Conduct a preliminary evaluation of proposal or response and shortlist the vendors.
- Conduct a detailed evaluation of the short-listed proposals.
- Award the contract to the best overall proposal.
- Implement the information systems technology. (6)

5.6 Rental✓
- Gives the organisation full use of the computer technology for an agreed period of time at an agreed monthly cost✓

Lease ✓
- Similar to rental, but normally over a longer period of time and used for assets of higher value✓

Purchase✓
- Borrow the capital or money needed to buy the computer technology and pay the loan back over a period of time✓ (6)

[30]

QUESTION 6

6.1 6.1.1 Product in a low-growth area which has a high market share

6.1.2 Product in a high-growth market in which the company has a large market share

6.1.3 Product in a low-growth market share in which the company has a low market share

6.1.4 Product in a high-growth market but with a low-market share

6.1.5 Means by which the organisations examine the position of an organisation in terms of its state or level of achievement of some business measures for example market position, financial position, technology position, et cetera.

6.1.6 Set of applications that support key business areas, functions or portfolios, together with cross-functional business processes, communication and collaboration needs
OR
Set of information systems applications that support key business areas in the organisation, as well as cross-functional business operation

(6 × 2) (12)

6.2

SYSTEMS PORTFOLIO	COMPANY PORTFOLIO
Set of information system applications that support key business areas in the organisation, as well as cross-functional business operations	Set or mix of products, brands and services offered by a company to its client or defined in terms of its major business functions

(2 × 2) (4)

6.3 6.3.1 High
6.3.2 Star
6.3.3 Dog
6.3.4 High
6.3.5 Market growth

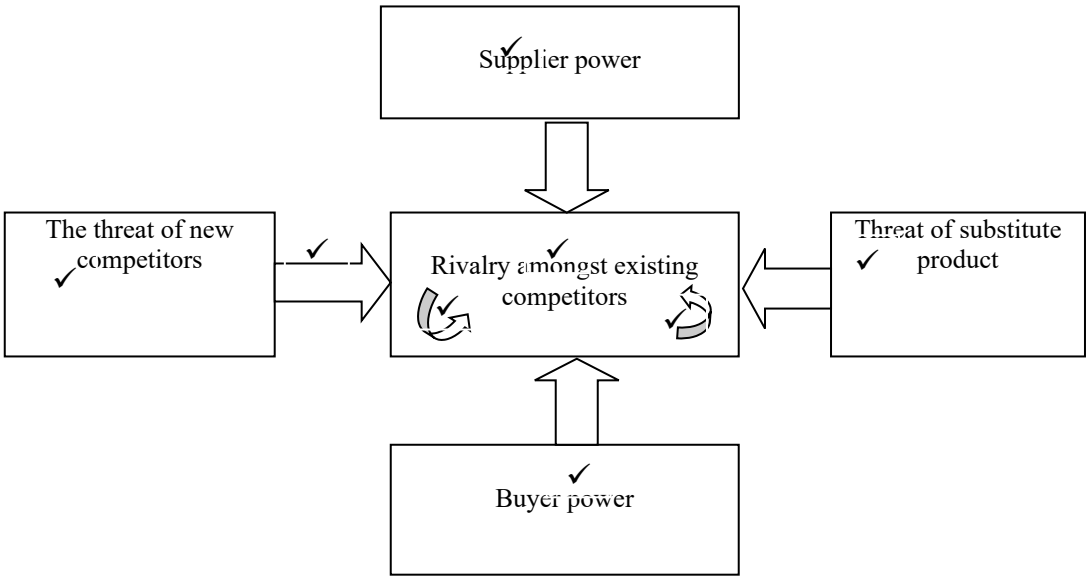
(5 × 1) (5)

6.4 6.4.1
- Information systems operations
- Information systems development
- Information systems support (3)

6.4.2
- Project identification
- Project evaluation
- Resource allocation (3)

6.4.3
- Business alignment
- Resource management
- Competitive advantage
- Organisational acceptance (4)

6.5



(8)
[39]

QUESTION 7

7.1
- Interview✓ – formal interaction between the systems analyst and the clients or systems users, conducted within an oral design✓
- Questionnaire✓ – documented set of questions formulated for gathering requirements and distributed to system users✓
- Focus group✓ – similar to interview, but conducted with a large number of system users and within a large room✓
- Observation and participation✓ – investigating skill through the use of contact view with the current system, thus observing and participating within the investigation of the current system✓
- Research into similar systems✓ – investigation similar systems that may be like the proposed information system✓ (5 × 2) (10)

7.2 Root cause analysis
- Focuses on the underlying causes of the problem and tries to remove them
- Used for more complex or serious problems
- Problems should be prioritised by users during problem analysis

Problem analysis
- Focuses on understanding the problem and finding a solution to it
- Seeks ideas and input from users and managers
- Useful and effective for improving efficiency and easy use of systems

OR
- Focuses on the problem, not the underlying problem (6)

7.3
- Identify all the potential costs and benefits.
- Allocate values or estimates to the costs and benefits.
- Estimate the cash flow of the project.
- Calculate the present net value of the project.
- Calculate the return on investment for the project.
- Determine the breakeven point for the project. (6)

7.4 7.4.1
A R3 090 000,00
B R6 250 000,00
C R310 000,00
D R1 700 000,00
E R5 500 000,00 (5)

7.4.2 $= \frac{\text{R9 050 000,00}\checkmark - \text{R8 090 000,00}\checkmark}{\text{R8 090 000,00}}$

$= 0{,}119\checkmark$

$= 11{,}9\%\checkmark$ (5)

[32]

TOTAL SECTION B: 160
GRAND TOTAL: 200

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